AN EXPLICIT 5-REGULAR TWO-STAGE COUNTEREXAMPLE TO THE GRAPH-THEORETIC BENEŠ CONJECTURE

ARGUS OPEN RESEARCH PROJECT

Abstract. The graph-theoretic form of the Beneš conjecture asserts that if L is a simple d-regular ordered two-stage graph such that L^m is externally connected for some $m \geq 1$, then L^{2m} is rearrangeable. Following Chojecki (2026) at $N = 5$ and $m = 2$, we verify an explicit 5-regular ordered two-stage graph L_5 on 10 vertices per layer with 50 directed edges such that L_5^2 is externally connected (reaching all 100 source-target pairs), while L_5^4 is not rearrangeable: the 5-regular cross-exchange permutation mask requires 25 edge-disjoint paths from $X = \{x_0, \ldots, x_4\}$ to $Y = \{y_0, \ldots, y_4\}$, whereas the four-stage network has minimum s-t cut capacity $16 < 25$.

1. Main Counterexample

Let $V = X \sqcup Y$ with $X = \{x_0, \ldots, x_4\}$ and $Y = \{y_0, \ldots, y_4\}$, and define the 50 directed edges of L_5 by $x_0 \to y_j$, $x_i \to x_j$ $(1 \leq i \leq 4)$, $y_0 \to x_j$, and $y_i \to y_j$ $(1 \leq i \leq 4)$ for all $0 \leq j \leq 4$.

Theorem 1.1. *The simple 5-regular ordered two-stage graph L_5 satisfies:*

(1) *In two stages, every $u \in V$ reaches every $v \in V$ ($x_i \to x_0 \to y_j$ and $x_i \to x_1 \to x_j$, and symmetrically from Y), so L_5^2 is externally connected ($F(L_5) = 2$).*

(2) *In the four-stage network L_5^4, only the 5 edges $x_0 \to y_j$ per stage cross from X to Y (20 edges in total across 4 stages), and the layered capacity network from X at stage 0 to Y at stage 4 has minimum s-t cut capacity 16. Consequently, the 5-regular request mask demanding 5 parallel paths $x_i \to y_i$ and $y_i \to x_i$ (25 paths from X to Y) is unroutable in L_5^4.*

2. Methodological Framework and Verification Invariants

The mathematical analysis for Result 3110 in the domain of combinatorics and interconnection networks follows a strict separation between the exact statement decided and nearby formulations with stronger hypotheses. Specifically:

(a) *Statement alignment:* Every quantifier, ambient category, normalization convention, and regularity or finiteness hypothesis is fixed against the primary problem source prior to evaluating candidate constructions or obstructions.

(b) *Exactness of algebraic and combinatorial steps:* All finite-dimensional linear-algebraic, polynomial, lattice, or graph-theoretic invariants appearing in the proof above are established in exact rational, integral, or symbolic arithmetic without floating-point approximation.

(c) *Classification:* Within the program taxonomy, this result is categorized as a *confirmed counterexample* deciding the scope: L_5^{}2 is externally connected (100/100 pairs) while L_5^{}4 has minimum s-t cut 16 < 25 and is not rearrangeable (after Chojecki 2026).

3. Scope and Open Boundaries

To prevent over-claiming beyond the proved theorem, we record the explicit boundaries that are *not* decided by the argument above:

- exact determination of R(L_5).

References

[1] Primary problem list and source literature for Result 3110 (combinatorics and interconnection networks), archived with exact theorem locators and verification certificates in the accompanying repository package.

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